KAIZEN MATHS

Lesson starter

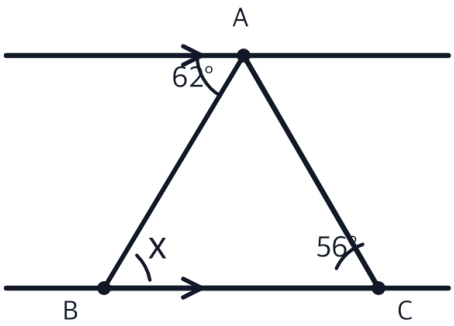
Formal Geometric Proof

Angle Chains

Name _____

SA1

Complete the proof chain using parallel-line facts and triangle angle facts.



The two horizontal lines are parallel. Prove that x = 62°.

GEO-EB04-L2 · SA

Show your working

KAIZEN MATHS

Lesson starter

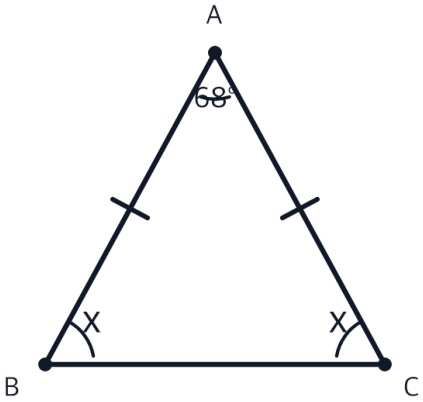
Formal Geometric Proof

Angle Chains

Name _____

SB1

Complete the proof chain using equal radii or equal sides.



AB = AC and angle A = 68°. Prove that each base angle is 56°.

GEO-EB04-L2 · SB

Show your working

KAIZEN MATHS

Lesson starter

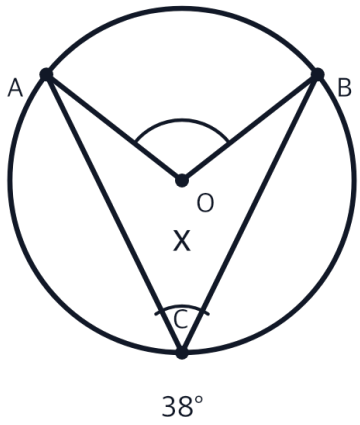
Formal Geometric Proof

Angle Chains

Name _____

SC1

Complete the proof chain using circle theorems.



Angle ACB = 38°. Prove that angle AOB = 76°.

GEO-EB04-L2 · SC

Show your working

KAIZEN MATHS

Lesson starter

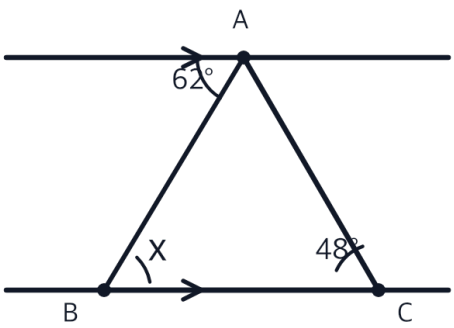
Formal Geometric Proof

Angle Chains

Name _____

SD1

Complete the proof chain using parallel-line facts and triangle angle facts.



The two horizontal lines are parallel. Prove that x = 70°.

GEO-EB04-L2 · SD

Show your working

KAIZEN MATHS

Exit ticket

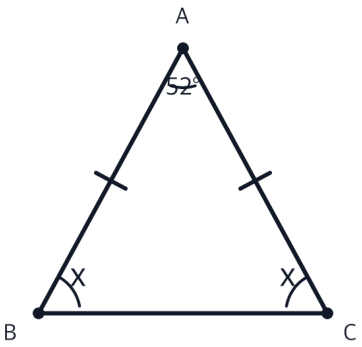
Formal Geometric Proof

Angle Chains

Name _____

EA1

Complete the proof chain using equal radii or equal sides.



AB = AC and angle A = 52°. Prove that each base angle is 64°.

GEO-EB04-L2 · EA

Show your working

KAIZEN MATHS

Exit ticket

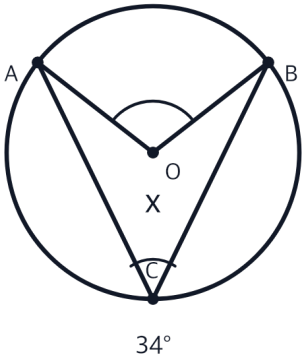
Formal Geometric Proof

Angle Chains

Name _____

EB1

Complete the proof chain using circle theorems.



Angle ACB = 34°. Prove that angle AOB = 68°.

GEO-EB04-L2 · EB

Show your working

KAIZEN MATHS

Exit ticket

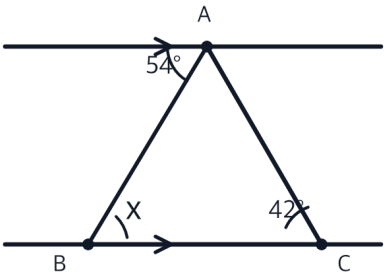
Formal Geometric Proof

Angle Chains

Name _____

EC1

Complete the proof chain using parallel-line facts and triangle angle facts.



The two horizontal lines are parallel. Prove that x = 84°.

GEO-EC04-L2 · EC

Show your working

KAIZEN MATHS

Exit ticket

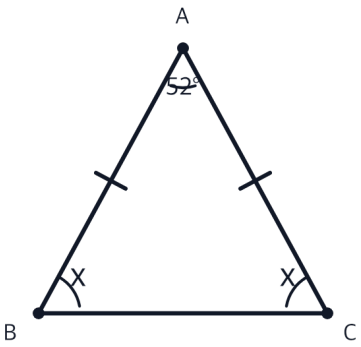
Formal Geometric Proof

Angle Chains

Name _____

EA1

Complete the proof chain using equal radii or equal sides.



AB = AC and angle A = 52°. Prove that each base angle is 64°.

GEO-EA04-L2 · EA

Show your working

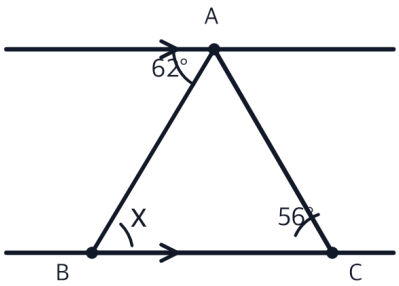
Teacher answers & working

Formal Geometric Proof · Angle Chains

GEO-EB04-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Complete the proof chain using parallel-line facts and triangle angle facts.



The two horizontal lines are parallel. Prove that x = 62°.

Answer:

62°

Working:

Statement	Reason
Angle at A in triangle ABC = 62°	Alternate interior angles are equal because the lines are parallel.
Angle at C = 56°	Given in the diagram.
$x + 62° + 56° = 180°$	Angles in a triangle add to 180°.
$x = 180° - 62° - 56°$	Rearrange to isolate x.
$x = 62°$	Calculate the value.

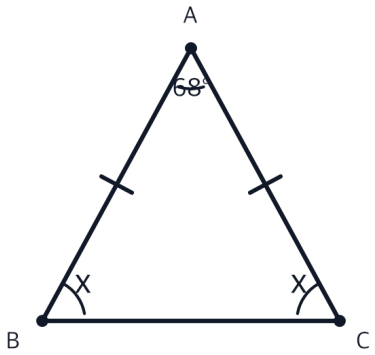
Teacher answers & working

Formal Geometric Proof · Angle Chains

GEO-EB04-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1

Complete the proof chain using equal radii or equal sides.



AB = AC and angle A = 68°. Prove that each base angle is 56°.

Answer:

56°

Working:

Statement	Reason
$AB = AC$	Given in the diagram.
Angle B = angle C	Base angles in an isosceles triangle are equal.
Angle A + angle B + angle C = 180°	Angles in a triangle add to 180°.
$68° + x + x = 180°$	Substitute angle A = 68° and angle B = angle C = x.
$2x = 112°$	Subtract 68° from both sides.
$x = 56°$	Divide by 2.

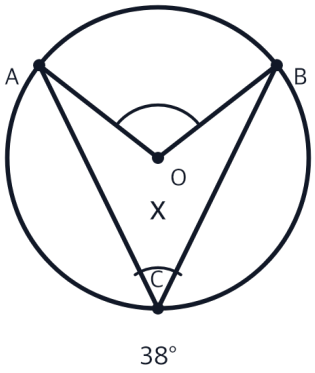
Teacher answers & working

Formal Geometric Proof · Angle Chains

GEO-EB04-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1

Complete the proof chain using circle theorems.



Angle ACB = 38°. Prove that angle AOB = 76°.

Answer:

76°

Working:

Statement	Reason
Angle ACB stands on arc AB.	A and B are the endpoints of the chord or arc.
Angle AOB also stands on arc AB.	O is the centre and OA, OB are radii.
Angle AOB = 2 × angle ACB	The angle at the centre is twice the angle at the circumference on the same arc.
Angle AOB = 2 × 38°	Substitute angle ACB = 38°.
Angle AOB = 76°	Calculate.

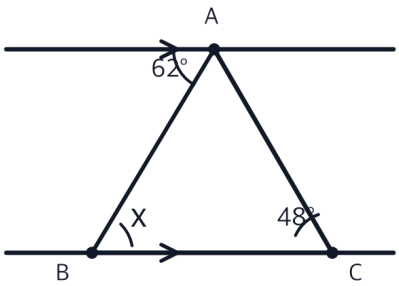
Teacher answers & working

Formal Geometric Proof · Angle Chains

GEO-EB04-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1

Complete the proof chain using parallel-line facts and triangle angle facts.



The two horizontal lines are parallel. Prove that x = 70°.

Answer:

70°

Working:

Statement	Reason
Angle at A in triangle ABC = 62°	Alternate interior angles are equal because the lines are parallel.
Angle at C = 48°	Given in the diagram.
$x + 62° + 48° = 180°$	Angles in a triangle add to 180°.
$x = 180° - 62° - 48°$	Rearrange to isolate x.
$x = 70°$	Calculate the value.

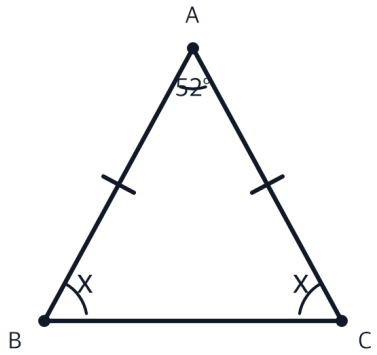
Teacher answers & working

Formal Geometric Proof · Angle Chains

GEO-EB04-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1

Complete the proof chain using equal radii or equal sides.



AB = AC and angle A = 52°. Prove that each base angle is 64°.

Answer:

64°

Working:

Statement	Reason
$AB = AC$	Given in the diagram.
Angle B = angle C	Base angles in an isosceles triangle are equal.
Angle A + angle B + angle C = 180°	Angles in a triangle add to 180°.
$52° + x + x = 180°$	Substitute angle A = 52° and angle B = angle C = x.
$2x = 128°$	Subtract 52° from both sides.
$x = 64°$	Divide by 2.

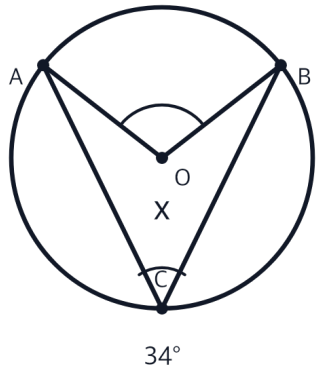
Teacher answers & working

Formal Geometric Proof · Angle Chains

GEO-EB04-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1

Complete the proof chain using circle theorems.



Angle ACB = 34°. Prove that angle AOB = 68°.

Answer:

68°

Working:

Statement	Reason
Angle ACB stands on arc AB.	A and B are the endpoints of the chord or arc.
Angle AOB also stands on arc AB.	O is the centre and OA, OB are radii.
Angle AOB = 2 × angle ACB	The angle at the centre is twice the angle at the circumference on the same arc.
Angle AOB = 2 × 34°	Substitute angle ACB = 34°.
Angle AOB = 68°	Calculate.

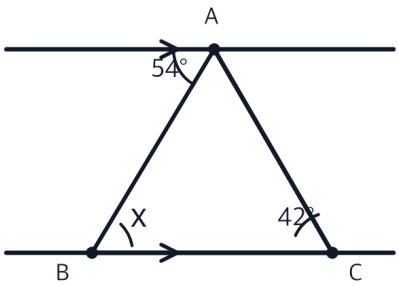
Teacher answers & working

Formal Geometric Proof · Angle Chains

GEO-EB04-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1

Complete the proof chain using parallel-line facts and triangle angle facts.



The two horizontal lines are parallel. Prove that x = 84°.

Answer:

84°

Working:

Statement	Reason
Angle at A in triangle ABC = 54°	Alternate interior angles are equal because the lines are parallel.
Angle at C = 42°	Given in the diagram.
$x + 54° + 42° = 180°$	Angles in a triangle add to 180°.
$x = 180° - 54° - 42°$	Rearrange to isolate x.
$x = 84°$	Calculate the value.